Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

If $A=\begin{bmatrix}1 & 2\\3 & 7\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}7 & -2\\-3 & 1\end{bmatrix}$, then calculate the inverse of A^{t} (if exists).

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

Evaluate the determinant $\begin{vmatrix} 2 & 4 & 8 & 16 \\ 0 & 3 & 6 & 12 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & -3 \end{vmatrix}$.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

If $A=\begin{bmatrix}1 & 2\\ 3 & 7\end{bmatrix}$ and $B=\begin{bmatrix}4 & 5\\ 3 & 4\end{bmatrix}$, then calculate the inverse of AB (if exists) ; where $A^{-1}=\begin{bmatrix}7 & -2\\ -3 & 1\end{bmatrix}$ and $B^{-1}=\begin{bmatrix}4 & -5\\ -3 & 4\end{bmatrix}$.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Marks: 3 (Budgeted Time 6 Min)

Let S be a parallelogram determined by the vectors $\vec{b}_1=\begin{bmatrix}-2\\3\end{bmatrix}$, $\vec{b}_2=\begin{bmatrix}-2\\5\end{bmatrix}$ and $A_1=\begin{bmatrix}6 & -2\\-3 & 2\end{bmatrix}$. Compute the area of the image of S under the mapping $\vec{x} \to A\vec{x}$.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Find an LU - decomposition of the matrix $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

 Marks: 5 (Budgeted Time 10 Min)

If $A=\begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & -3 \\ 9 & -3 & 3 \end{bmatrix}$, then find the minors of first row and determinant of A using these minors .

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

If a matrix A is decomposed into $L = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & -7 \\ 0 & 1 \end{bmatrix}$ by LU factorization , then find A.

Answer (Please click here to Add Answer)

100%

Normal Arial 12

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

If $A=\begin{bmatrix}4 & 2 & 3\\ 1 & 3 & 2\end{bmatrix}$, then determine whether $\vec{u}=\begin{bmatrix}1\\ 4\\ 5\end{bmatrix}$ is in $Nul\ A$ or not ?

Answer (Please click here to Add Answer)

100%

Normal Arial 12

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Let W be the set of all vectors of the form $\begin{bmatrix} 3a \\ 2a \\ a \end{bmatrix}$. Find a vector $\vec{v}$ in R^3 such that $W = Span\{\vec{v}\}$.

Answer (Please click here to Add Answer)

100%

Normal Arial 12

Construct partitions of the following matrix into three 2×2 blocks :

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 6 & 3 & 4 \end{bmatrix}$$

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Show that the coefficient matrix of the following linear system is strictly diagonally dominant. Also calculate the second iteration using Gauss – Seidel method if the first iteration is $(x_1^{(1)}, x_2^{(1)}, x_3^{(1)}) = (1.3, 1.04, 0.936)$.

$$10x_1 + 2x_2 + x_3 = 13$$
$$2x_1 + 10x_2 + x_3 = 13$$
$$2x_1 + x_2 + 10x_3 = 13$$

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Show that $\begin{vmatrix} a+h & a-h & d \\ b+k & b-k & e \\ c+l & c-l & f \end{vmatrix} = 2\begin{vmatrix} a & a-h & d \\ b & b-k & e \\ c & c-l & f \end{vmatrix}$.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

If a matrix A is decomposed into $L = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & -7 \\ 0 & 1 \end{bmatrix}$ by LU factorization , then find A.

Answer (Please click here to Add Answer)

100%

Normal Arial 12

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

If $A = \begin{bmatrix} 4 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}$, then determine whether $\vec{u} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$ is in $Nul\ A$ or not ?

Answer (Please click here to Add Answer)

Normal Arial 12 100%

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Normal Arial 12

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Answer (Please click here to Add Answer)

100%

Normal Arial 12

 Marks: 2 (Budgeted Time 4 Min)

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$, then calculate the inverse of A^t (if exists).

Answer (Please click here to Add Answer)

Normal Arial 12 100%

 Marks: 2 (Budgeted Time 4 Min)

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Normal Arial 12 100%

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Answer (Please click here to Add Answer)

Normal Arial 12 100%

 Marks: 3 (Budgeted Time 6 Min)

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Normal Arial 12 100%

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